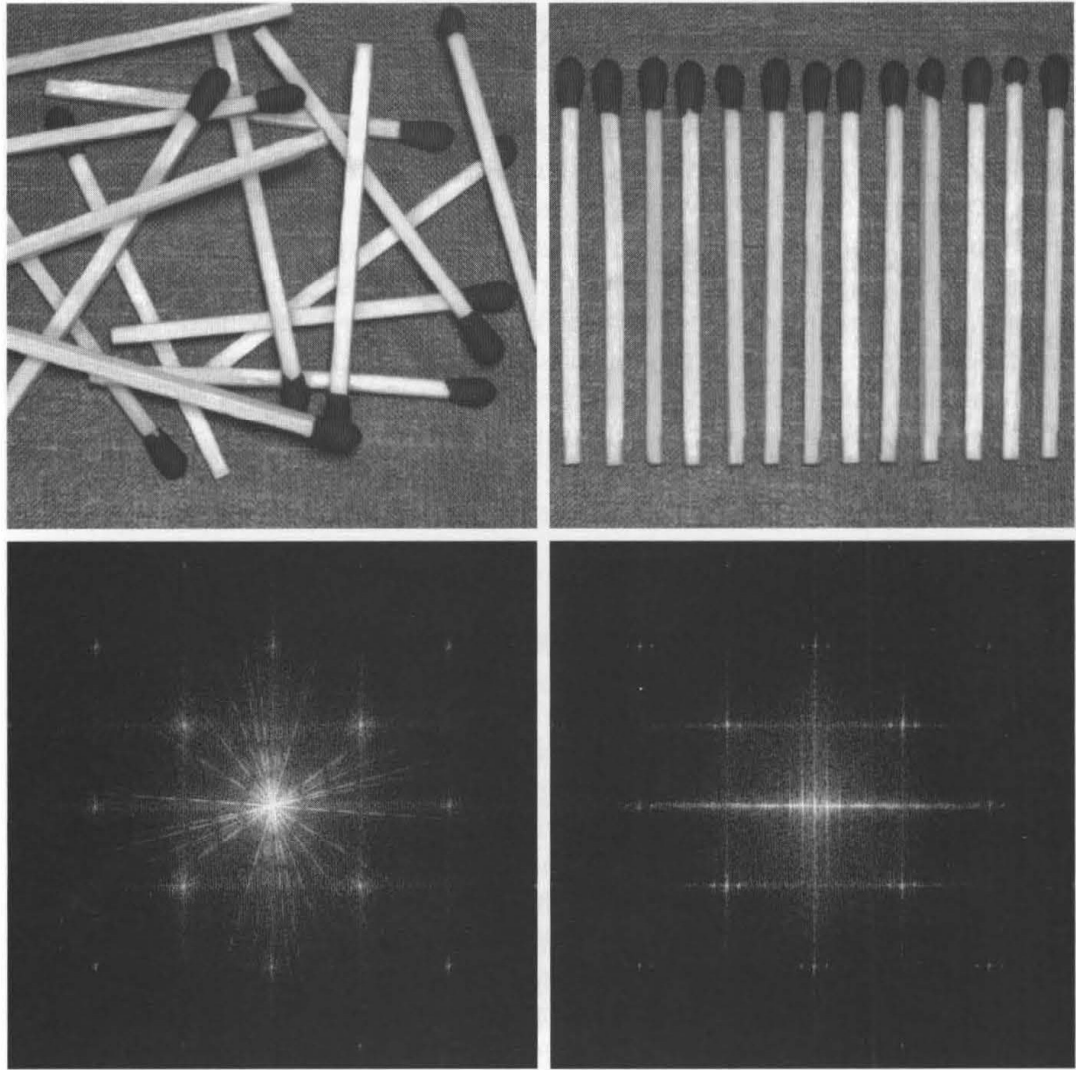


a	b
c	d

FIGURE 12.26

(a) and (b)
Images of
unordered and
ordered objects.
(c) and (d)
Corresponding
spectra.



12.4.3 Moment Invariants

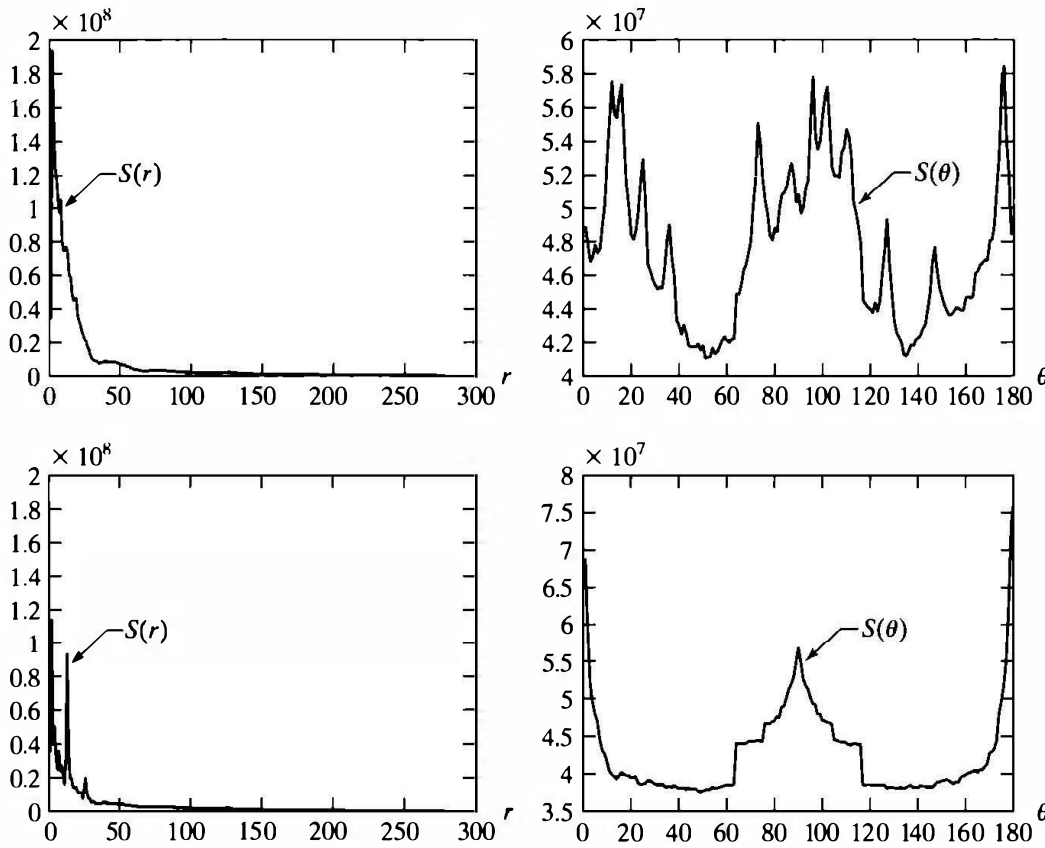
The 2-D *moment* of order $(p + q)$ of a digital image $f(x, y)$ of size $M \times N$ is defined as

$$m_{pq} = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} x^p y^q f(x, y)$$

where $p = 0, 1, 2, \dots$ and $q = 0, 1, 2, \dots$ are integers. The corresponding *central moment* of order $(p + q)$ is defined as

$$\mu_{pq} = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} (x - \bar{x})^p (y - \bar{y})^q f(x, y)$$

for $p = 0, 1, 2, \dots$ and $q = 0, 1, 2, \dots$, where



a	b
c	d

FIGURE 12.27
(a) and (b) Plots of $S(r)$ and $S(\theta)$ for the random image in Fig. 12.26(a). (c) and (d) Plots of $S(r)$ and $S(\theta)$ for the ordered image.

$$\bar{x} = \frac{m_{10}}{m_{00}} \quad \text{and} \quad \bar{y} = \frac{m_{01}}{m_{00}}$$

The *normalized central moment* of order $(p + q)$ is defined as

$$\eta_{pq} = \frac{\mu_{pq}}{\mu_{00}^\gamma}$$

where

$$\gamma = \frac{p + q}{2} + 1$$

for $p + q = 2, 3, \dots$

A set of seven 2-D *moment invariants* that are insensitive to translation, scale change, mirroring (to within a minus sign), and rotation can be derived from these equations.[†] They are listed in Table 12.6.

[†] Derivation of these results involves concepts that are beyond the scope of this discussion. The book by Bell [1965] and a paper by Hu [1962] contain detailed discussions of these concepts. For generating moment invariants of order higher than seven, see Flusser [2000]. Moment invariants can be generalized to n dimensions (see Mamistvalov [1998]).

TABLE 12.6

A set of seven
moment
invariants.

Moment order	Expression	Too much Important
1	$\phi_1 = \eta_{20} + \eta_{02}$	
2	$\phi_2 = (\eta_{20} - \eta_{02})^2 + 4\eta_{11}^2$	
3	$\phi_3 = (\eta_{30} - 3\eta_{12})^2 + (3\eta_{21} - \eta_{03})^2$	
4	$\phi_4 = (\eta_{30} + \eta_{12})^2 + (\eta_{21} + \eta_{03})^2$	
5	$\phi_5 = (\eta_{30} - 3\eta_{12})(\eta_{30} + \eta_{12})[(\eta_{30} + \eta_{12})^2 - 3(\eta_{21} + \eta_{03})^2] + (3\eta_{21} - \eta_{03})(\eta_{21} + \eta_{03})[3(\eta_{30} + \eta_{12})^2 - (\eta_{21} + \eta_{03})^2]$	
6	$\phi_6 = (\eta_{20} - \eta_{02})[(\eta_{30} + \eta_{12})^2 - (\eta_{21} + \eta_{03})^2] + 4\eta_{11}(\eta_{30} + \eta_{12})(\eta_{21} + \eta_{03})$	
7	$\phi_7 = (3\eta_{21} - \eta_{03})(\eta_{30} + \eta_{12})[(\eta_{30} + \eta_{12})^2 - 3(\eta_{21} + \eta_{03})^2] + (3\eta_{21} - \eta_{03})(\eta_{21} + \eta_{03})[3(\eta_{30} + \eta_{12})^2 - (\eta_{21} + \eta_{03})^2]$	

Custom M-function `invmoments` implements these seven equations. The syntax is as follows (see Appendix C for the code):

```
invmoments
```

```
phi = invmoments(f)
```

where `f` is the input image and `phi` is a seven-element row vector containing the moment invariants just defined.

EXAMPLE 12.13:
Moment
invariants.

■ The image in Fig. 12.28(a) was obtained from an original of size 400×400 pixels using the following commands:

```
>> f = imread('Fig1228(a).tif');
>> fp = padarray(f, [84 84], 'both'); % Padded for display.
```

This image was created using zero padding to make all displayed images consistent in size with the image occupying the largest area (568×568) which, as explained below, is the image rotated by 45° . The padding is for display purposes only, and was not used in moment computations. A translated image was created using the following commands: